

Ali Kuwajerwala

Researcher and Engineer, AI and Robotics

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AI and robotics researcher and engineer with a strong background in deep learning, robotics, and multimodal foundation models. Seeking technical research and engineering roles.

SELECTED PUBLICATIONS

- ICRA 2024** *ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning*
Kuwajerwala, A., Gu, Q., Morin, S., et al. <https://concept-graphs.github.io>.
- RSS 2023** *ConceptFusion: Open-set Multimodal 3D Mapping*
Jatavallabhula, K. M., Kuwajerwala, A., Gu, Q., et al. <https://concept-fusion.github.io>.

EXPERIENCE

Contract Research Engineer Oct. 2025 – Present
Multiple Stealth AI & Robotics Startups Remote

- Built policy-training infrastructure for reinforcement-learning workflows in simulation, targeting whole-body robot control and sim-to-real transfer.
- Built data pipelines for whole-body motion imitation, ingesting and normalizing large motion-capture datasets, filtering low-quality clips, and exporting training-ready datasets and data loaders.
- Developed perception and mapping software for early-stage mobile robot prototypes, including SLAM integration, sensor-processing pipelines, and field-calibration workflows.

Member of Technical Staff, K-Scale Labs (Startup) Nov. 2024 - Sep. 2025
Founding Team, Deep Learning (Palo Alto, CA)

- Led humanoid robot sim-to-real locomotion on IsaacLab², implementing rewards, actuator models, curriculum, domain randomization, logging and evaluation pipelines.
- Created K-Scale MuJoCo Viewer (KMOV)², an interactive simulation visualizer. Built with **Qt** and **Python**, enables rapid debugging with push forces, real-time plotting, logging, and video saving.
- Created K-Infer Evals², an automation system for evaluating trained locomotion policies. Runs policies in a variety of scenarios and logs results to a Notion database for comparison and inspection.
- Core contributor to the CAD → URDF (pipeline²) with inertia calibration & consistency checks to keep mechanical design, simulation, and firmware in sync.

Applied Scientist Intern, Amazon Summer 2022
Alexa AI Team, Amazon Devices (Toronto, ON)

- Improved the accuracy of the conversational **NL2SQL** system by **1.5%** on the **Spider NL2SQL** dataset.
- Prototyped alternative model architectures to overcome the 512 token length limitation in existing models.

Machine Learning Engineer, Liquid Analytics (Startup) Summer 2021
Perform AI Application, Core Algorithms Team (Remote, US)

- Developed highly scalable algorithms in **Julia** to quickly process logistics data for large distribution companies.
- Set up queuing infrastructure using **AMQP** and **RabbitMQ** to handle upto 300,000 requests each second.

EDUCATION

Mila & University of Montréal
M.Sc., Computer Science (Robotics and Artificial Intelligence) Sep. 2021 – Aug. 2024

- Supervisor:** Prof. Liam Paull, director of the Montreal Robotics and Embodied AI Lab, Core Mila Member, CIFAR AI Chair.

University of Toronto
H.B.Sc., Computer Science & Math CGPA: 3.63 Sep. 2016 – May 2020

- Award:** Received the NSERC Undergraduate Student Research Award, a value of **\$5600**. (2020)
- Extracurricular:** Co-Founder & Head of Operations of the Robotics Club. (2019-2020)
- Teaching Assistant:** Mobile Robotics (CSC477), Data Structures (CSC263), Theory of Computation (CSC236).

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, Julia, SQL
Libraries / Tools: PyTorch, JAX, Tensorflow, ROS